Generator–coefficient comparison: all 21 curves

29 October 2026. The declared finite comparison completed on ten published and eleven omitted curves across all six field sizes, including P-384 and P-521. The separate native verifier checked every declared row. None of the 19,152 single published-envelope tests separates all labels, and none of the 880 fixed public-scalar checks matches a standardized generator. These negative results exclude only the stated families. The historical skip criterion and both weakness hypotheses remain unresolved.

Verified observations

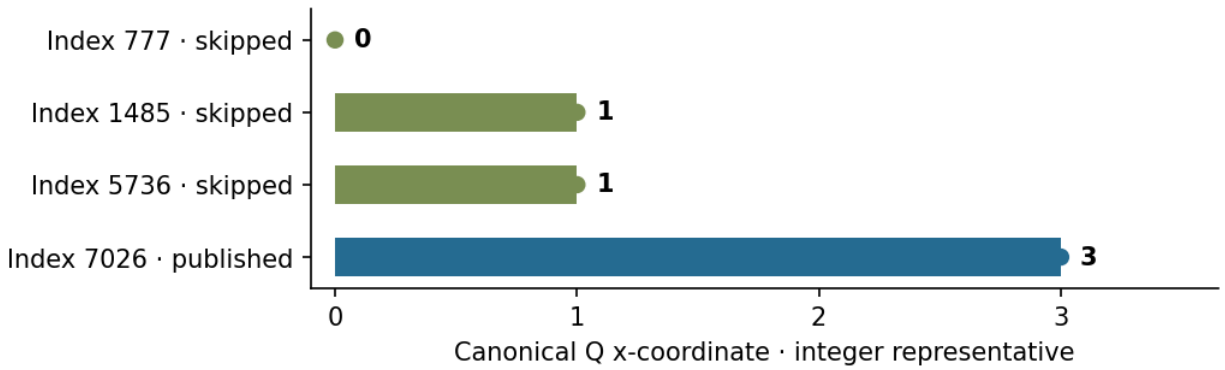
Recorded quantity	Actual result, attempt 002
Original choices	21: ten published, eleven omitted; zero excluded
Repeated representations	Two coefficient views; fourteen point rules; 588 point rows
Point/coefficient traits	171 per point; 100,548 cells independently checked
Fixed single envelopes	19,152 checked; zero exact separators
Same-field differences	119,700 point-trait cells plus 28,175 inherited-feature cells
Known-scalar equality checks	880 checked; zero matches
Historical generators of omitted curves	All eleven remain literal NA

Sources: native summary, separate verification, display data and canonical export. These are deterministic counts on a fixed census, with no sampling interval or inferential p-value. Repeated representations are controls on the same curves, not additional historical samples.

A trait that stands out in P-224

For Q chosen by the user's first-x, smaller-y rule, the three omitted P-224 curves have x=0, 1, 1 at indices 777, 1485, 5736. The published index 7026 has x=3. Both explicit coefficient views give those values. Q is an assigned canonical point; this does not say that the standardized published G has x=3. All seventeen non-P-224 canonical x values are zero in both label groups.

The predeclared per-field lower envelope for x:bits, k=1, sign=+1 keeps all ten publications and rejects the three P-224 omissions, but retains eight other omissions: p192 indices 2198/5405, p239 indices 36/1562/2491/2682/2863, and p256 index 876. Thus this narrow pattern is not a full-cohort criterion. Selecting it for discussion after viewing the results does not establish historical use, statistical significance or vulnerability.



Attempt 002 · 4 P-224 choices · both coefficient views agree · exact values, no sampling interval

What connects a point to b

For a=−3, an on-curve point satisfies b = $y^2 - x^3 + 3x$ (mod p). Consequently x=0 implies y^2=b and x=1 implies y^2=b−2. These explain the close G–b relations of the four user-supplied assigned points. Conditional on the generator equation $c*b^2=-27$, they also give $c*y^4=-27$ at x=0 and $c*(y^2+2)^2=-27$ at x=1. These are defining-equation consequences. The five previously checked conditional Lean theorems and their precise assumptions are in the proof map. They are not formal proofs of historical selection or the full isomorphism.

The inherited feature records have square b for 9/10 publications and 9/11 omissions. All seventeen non-P-224 b values are square: under the reference convention p≡7 mod8 and b=R^((p+1)/4), the exponent is even. That construction forces the canonical search to start at x=0 for both labels. P-224 index 777 is square; indices 1485, 5736 and published 7026 are nonsquare. These b-character counts reuse the earlier proved-factor study and are not newly recomputed factor families.

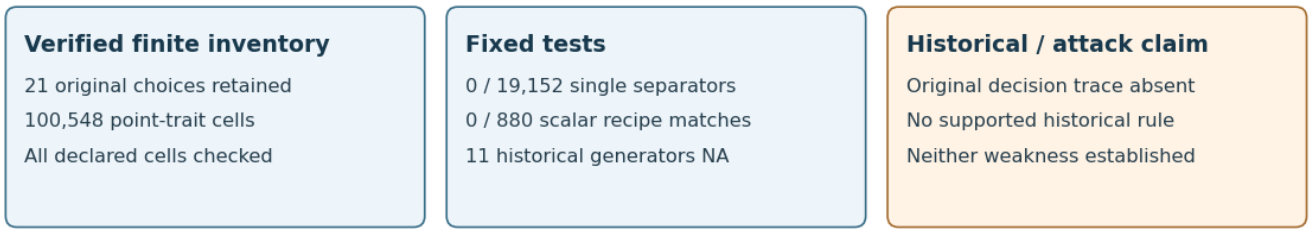
Matched controls and the tested family

The historical-b view and smaller-b P-224 view retain exact model identities. For P-224, b→−b is transported by (x,y)→(−x,i*y), $i^2=-1$ in Fp. Native checks validate the actual transported points and orders. This same-order identification is not applied to p≡3 mod4 twists. The four user-supplied canonical points are checked exactly in their stated views.

Each view evaluates ±[k]Q for k in {1,2,3,5,7,11,17}. The 171 traits include coordinate bit lengths, Hamming weights, modular distances, quadratic characters, small residues and a fixed polynomial-equality catalog. For each trait and point rule, global and per-field tight upper and lower envelopes retain every published control. Every same-field published/omitted pair is retained. No new conjunction was fitted here. The inherited 1,127 features are hash-bound copies from the complete 84-factor-record study: their pairwise differences are independently checked, but their original factorizations are not rerun in this addition.

The 22 fixed scalar recipes use b, nearby powers/offsets, seed/hash values, bit slices, trace, index/run and small constants. Ten known standardized generators × two views × 22 recipes × two directions give 880 checks: G=±[s]Q and Q=±[s]G. Signs are included inside each equality; these tests use Q, not every multiple control. The exact recipe definitions and encodings are in the frozen family and native source. A known-scalar test is not DLP recovery.

Earlier reconstruction already links six May-1997 published base points to the FreeLIP RNG stream. The present zero matches do not refute that different, previously reported relation. See the FreeLIP source discussion.



Actual attempt 002 · Controls are repeated representations, not new historical samples

Execution, failures and admission

The primary computation uses native C++/PARI; a separately implemented GMP affine verifier recomputes the group/trait operations, cutoffs, differences, recipes and summary. Both share GMP integer primitives and OpenSSL SHA. The earlier tiny-fixture integration and nine corruption refusals remain development evidence, not historical-curve measurements.

Flow	Actual phases and outcome
Attempt 001, frozen protocol 1	Three planned; two attempted; one completed; production timed out (124); verification not attempted; none excluded
Resource diagnostics	Four primitive invocations and one all-input invocation; two timeouts and three successes; no point-trait comparison
Attempt 002, frozen protocol 2	Three planned, attempted and completed; all statuses zero; no interruption, resume or exclusion

Attempt 001 wrote zero point-trait rows before its 300-second timeout. Diagnostics localized the stall to proof of the fixed known P-192 field prime under a 768 MiB virtual-address ceiling. The exact original diagnostic binary succeeds at 1,536 MiB; all 21 known p/n pairs, covering six field sizes, then pass the separate resource check. An APRCL diagnostic at the smaller ceiling also timed out; it was not substituted into production.

Protocol 2 explicitly raises only the uniform address ceiling to 1,536 MiB. It preserves the original binaries, prover, all inputs and comparisons, and 30/300/300-second phase limits. It was frozen and committed before attempt 002. This is an execution repair, not evidence of weak curves. The freeze digest is 090d27110aa85eac1309733bdb9521244a7e4bb6ad9d624cd4d34fcc94eb9e85; the actual launch source is 1b18bec823d730ef8f30ef0761595cac633c411e. Frozen plan 2 retains two stale descriptive path strings; the actual bound launcher is run-v2.sh. The additive correction preserves both the discrepancy and the executed script without rewriting it.

The canonical experiment-protocol tool at revision 8c5db250d4fff65eb40fffede9c8391de403897f admits version 2 with file verification and zero findings. Version 1's partial export is refused with four findings because verification never ran. Its adapter's flow describes the two existing rows; the actual planned count remains three. No cancelled verification attempt or complete census was manufactured. The export limitation remains explicit and does not invalidate the complete version-2 execution.

Execution used the declared local exact-analysis fallback. The portable EKS request contract could not express this native job; no EKS MCP was available and CodeBuild project inspection was denied. No paid job or administrative policy change was substituted. Previously authorized access authenticated and uploaded the required checkpoint before each run. The earlier restored runtime access block was corrected, not retained as the current status. No performance clock or isolation level is claimed for the actual comparison.

Run reports and PDFs: 001, 002, diagnostics, synthetic preflight.

Requested work and remaining uncertainty

User request	Status and evidence
Original four ICV1/DiSSECT assessments	Previously delivered in the original study
Complete four unresolved factorizations and compare	Verified complete: 84/84 proved records on all 21 curves; no unknown factor values used as labels or silently omitted
Wider differences and G–b tests	Verified complete for this declared finite family; 100,548 point-trait cells and all 147,875 difference cells checked
Explain the historical skips	Unresolved; no original caller, acceptance or visitation records; no independent historical holdout
Weak published or weak skipped curves	Neither established; an applicable attack construction and validation remain missing
PDF and PR	This current PDF and existing PR 2666

Earlier structural conjunctions remain post-hoc fits; zero single separators in this different point-trait family does not erase those fits. This is not an exhaustive search over mathematical functions or all historical programs. Eligible stream indices do not prove that the historical program visited and rejected each curve. The user has no original decision logs. Distinguishing the two weakness hypotheses still requires a specified attack contract, applicable construction and independently checked outcome. Trait differences alone supply none of these. No DLP, security proof or security break is claimed. Modern input prime checks do not repair the twelve indeterminate legacy ECPP certificate rows. They remain unknown in their own study.

Custody, reproduction and sources

The preparation archive, both actual attempts, synthetic planning outputs and all resource diagnostics are retained. Full remote GET, byte/SHA checks and internal member checks passed for all four final native archives; the preparation has its own verified receipt. The full raw bundle is evidence/full-raw-002.tar.xz, 935,468 bytes, SHA-256 d15967a549fea14af11733b5685cad6abed9291f3c1a78820e21e3c15a2aa909, with 106 regular members. The receipt is final-native-delivery.json. REPRODUCE.md explains hydration and the canonical file check. Native libraries are pinned, not bundled. Credentials are excluded.

Primary methods reviewed before implementation: SEC 1 v2 §3.1.3.2 describes seed/hash point generation; RFC 5639 A.2 provides a public scalar multiple of a smallest-x point. These motivate comparison families, not attribution to the 1997 caller. Published generators come from std-curves revision 77fe6e3585ca2c2225b59d7df24b7c775437276f. The PARI arithmetic documentation informs the known-prime proof diagnostics; no probable-prime substitution was made. The curve-origin account, source recheck, FreeLIP trial and proved-factor study bound novelty and the remaining historical gap. No original acceptance log was found or supplied.

The new canonical-x graph and execution diagram supersede the preparation's unrun status additively. Its old figures remain frozen planning evidence. Factor, embedding, FreeLIP and ECPP-work graphs retain their earlier values. IC leaderboard, progress timeline and performance-gain charts are unchanged: this study measures coverage and exact traits, not an IC improvement. Synthetic planning times have browsing copies outside the admitted benchmark table; they lack performance validation and a matched rho reference. Cargo, Conductor and lean4checker are unavailable; no whole-repository pass is claimed.